Shaunak HW9

1. if ρ is the correlation coeff^n for X and Y,

to show, $|\rho| \leq 1$.

$$\rho = \frac{Cov(X, Y)}{\sigma_X \sigma_Y}$$

so $|Cov(X, Y)| \leq \sigma_X \sigma_Y$ to be proved.

$$|E[(X - \mu_X)(Y - \mu_Y)]| \leq \sigma_X \sigma_Y$$

using Cauchy-Schwarz inequality

$$|E[(X - \mu_X)(Y - \mu_Y)]|^2 \leq E[(X - \mu_X)^2 . E[(Y - \mu_Y)^2]$$

$$\therefore \quad Cov(X, Y)^2 \leq \sigma_X^2 \sigma_Y^2$$

taking square root

$$|Cov(X, Y)| \leq \sigma_X \sigma_Y$$

$$\therefore \quad \frac{|Cov(X, Y)|}{\sigma_X \sigma_Y} \leq 1.$$

$$\therefore \quad |\rho| \leq 1$$

2. X and Y are uncorrelated random variables.

to show, $\sigma_{X+Y}^2 = \sigma_X^2 + \sigma_Y^2$

if $\rho = 0$, then X & Y are uncorrelated.

$\therefore \quad Cov(X, Y) = 0$

$$\sigma^2_{X+Y} = E\left[\left(X + Y - \mu_X - \mu_Y\right)^2\right]$$

$$= E\left[\left((X - \mu_X) + (Y - \mu_X)\right)^2\right]$$

$$= E\left[(X - \mu_X)^2\right] + E\left[(Y - \mu_Y)^2\right] + 2 \cdot E\left[(X - \mu_X)(Y - \mu_Y)\right]$$

$$= \sigma_X^2 + \sigma_Y^2 + 2 C_{XY} \qquad \because C_{XY} = 0$$

$$= \sigma_X^2 + \sigma_Y^2$$

Q.E.D.

(5) X and Y are equal in mean square sense if $E[(X-Y)^2] = 0$

$\rho = 1$ given so $\dfrac{Cov(X, Y)}{\sigma_X \sigma_Y} = 1$.

$$\therefore \quad Cov(X, Y) = \sigma_X \cdot \sigma_Y$$

I followed the solution posted to answer this one.

we need to find a, b & here

$$E\left[(Y - (aX + b))^2\right] \geq 0 \quad \forall\, a \,\&\, b$$

so we need to minimize $E\left[(Y - (aX + b))^2\right]$

Differentiating wrt a & b

$$\frac{\partial}{\partial a}\left[E\left((Y - (aX + b))^2\right)\right] = 0 \implies E\left[\frac{\partial}{\partial a}\left(Y - (aX + b)^2\right)\right] = 0 \quad \text{—— (1)}$$

$$\& \quad E\left[\frac{\partial}{\partial b} E\left(Y - (aX + b)\right)^2\right] = 0 \quad \text{—— (2)}$$

from 1 $\quad E[-2X(Y - aX - b)] = 0$

$$E[XY - aX^2 - bX] = 0$$

$$E[XY] - aE[X^2] - bE[X] = 0 \quad \text{—— (3)}$$

from 2 $\quad E[-2(Y - aX - b)] = 0$

$$E[Y] - aE[X] - b = 0$$

$$\therefore \mu_Y - a\mu_X = b \quad \text{—— (4)}$$

4 in 3 $\quad E[XY] - aE[X^2] - (\mu_Y - a\mu_X)\mu_X = 0$

$$E[XY] - aE[X^2] - \mu_X\mu_Y + a\mu_X^2 = 0$$

$$E[XY] - \mu_Y\mu_X = a(E[X^2] - \mu_X^2)$$

$$C_{XY} = a\sigma_X^2 \quad \text{or } a = \frac{C_{XY}}{\sigma_X^2}$$

$$b = \mu_Y - \frac{C_{XY}}{\sigma_X^2}\mu_X, \text{ from 4}$$

$$\therefore Y = \frac{C_{XY}}{\sigma_X^2}X + \mu_Y - \frac{C_{XY}}{\sigma_X^2}\mu_X$$ in the mean sq sense

(4) 4 zero mean, unit variance r.v's T, U, V and W pairwise uncorrelated.

X, Y, Z such that $X = T + U$, $Y = U + V$, $Z = V + W$

$\therefore E[T] = E[U] = E[V] = E[W] = 0$ —— (1)

& $Var(T) = Var(U) = Var(V) = Var(W) = 1$. —— (2)

also $Cov(T, U) = Cov(T, V) = \ldots = Cov(V, W) = 0$ since pairwise uncorrelated.

from (2) since $Var(T) = E[T^2] - (E[T])^2$

but $E[T] = 0$ $\therefore Var(T) = E[T^2]$

$\therefore E[T^2] = E[U^2] = E[V^2] = E[W^2] = 1$.

next $E[X] = E[T + U] = E[T] + E[U] = 0$

$E[Y] = E[U + V] = E[U] + E[V] = 0$

$E[Z] = E[V + W] = E[V] + E[W] = 0$

$\therefore Var(X) = E[X^2] - (E[X])^2 = E[X^2] = E[(T+U)^2]$

$= E[T^2] + 2E[TU] + E[U^2]$

$= 1 + 2.0 + 1$

$= 2$

$Cov(T, U) = 0$ or $E[TU] - E[T] \cdot E[U] = 0$

$\therefore E[TU] = 0$

$\therefore Var(Y) = E[Y^2] = E[U^2] + 2.E[UV] + E[V^2]$

$= 1 + 0 + 1 = 2$

$Var(Z) = E[Z^2] = E[V^2] + 2E[VW] + E[W^2]$

$= 1 + 0 + 1 = 2$.

$\therefore$ Standard dev σ_X, σ_Y & σ_Z are $\sqrt{2}$ each case.

$$\therefore \rho_{XY} = \frac{Cov(X,Y)}{\sigma_X \sigma_Y}$$

$$Cov(X,Y) = E[(X-\mu_X)(Y-\mu_Y)]$$

$$= E[XY + \mu_X\mu_Y - X\mu_Y - Y\mu_X]$$

$$= E[XY] - E[X]E[Y]$$

$$= E[XY] - 0.$$

$$\therefore = E[(T+U)(U+V)]$$

$$= E[(TU + TV + U^2 + UV)]$$

$$= E[TU] + E[TV] + E[U^2] + E[UV]$$

$$= 0 + 0 + 1 + 0 = 1$$

$$\therefore \rho_{XY} = \frac{1}{\sqrt{2}.\sqrt{2}} = \frac{1}{2}$$

$$\rho_{XZ} = \frac{Cov(X,Z)}{\sigma_X \sigma_Z} = \frac{E[XZ]}{\sigma_X \sigma_Z} = \frac{E[(T+U)(V+W)]}{\sqrt{2}.\sqrt{2}}$$

$$= \frac{E[TV] + E[TW] + E[UV] + E[UW]}{2}$$

$$= \frac{0}{2} = 0$$

$\therefore$ Corr Coeff b/w X & Y is 0.5

" " " X & Z is 0.

(5) X and Y independent random variable with variances σ_X^2 and σ_Y^2. (6)

$$Z = aX + (1-a)Y \quad \text{where} \quad 0 \le a \le 1.$$

To find a that minimizes variance of Z

$$Var(Z) = Var\left(aX + (1-a)Y\right)$$

$$= Var(aX + bY) \quad \text{where } b = 1-a$$

$$= E\left[\left((aX+bY) - (aE[X] + bE[Y])\right)^2\right]$$

$$= E\left[\left(a(X - E[X]) + b(Y - E[Y])\right)^2\right]$$

$$= E\left[a^2(X - E[X])^2 + b^2(Y - E[Y])^2 + 2ab(X - E[X])(Y - E[Y])\right]$$

$$= a^2 E[(X - E[X])^2] + b^2 E[(Y - E[Y])^2] + 2ab E[(X - E[X])(Y - E[Y])]$$

Since X & Y are independent,

$$= a^2 Var(X) + b^2 Var(Y) + 2ab\, E[(X - E[X)].\, E[Y - E[Y]]$$

$$= a^2 Var(X) + b^2 Var(Y) + 0$$

$$= a^2 \sigma_X^2 + (1-a)^2 \sigma_Y^2 \qquad 0 \le a \le 1$$

$$\frac{\partial \sigma_Z^2}{\partial a} = 2a\sigma_X^2 + 2(1-a)(-1)\sigma_Y^2 = 0$$

$$\therefore 2a\sigma_X^2 - 2(1-a)\sigma_Y^2 = 0 \quad \text{or } a(\sigma_X^2 + \sigma_Y^2) = \sigma_Y^2$$

$$a = \frac{\sigma_Y^2}{\sigma_X^2 + \sigma^2} < 1 \quad \text{minimizes}$$

- jointly distributed r.v X and Y
- Independent iff $E[u(a-X)u(b-Y)] = E[u(a-X)]\,E[u(b-Y)]$

for all real numbers a & b, where $u(x)$ is unit step f^n.

X, Y independent if and only if $f(x,y) = f_X(x)\, f_Y(y)$

From definition of expectation,

$$E[X] = \int_{-\infty}^{\infty} x\, f_X(x)\,dx$$

$$E[g(X,Y)] = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} g(x,y)\, f_{XY}(x,y)\,dx\,dy$$

for product of RV's X and Y

$$E[X.Y] = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} xy.\, f_{XY}(x,y)\,dx\,dy$$

$$E[XY] = E[X].E[Y]$$ since X & Y are independent

let $X \equiv u(a-X)$ & $Y \equiv u(b-Y)$

To show $E[u(a-X)\,u(b-Y)] = E[u(a-X)]\,E[u(b-Y)]$

LHS

$$\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} 1_{(-\infty,a]}(a-X)\cdot 1_{(-\infty,b]}(b-Y)\,dx\,dy$$

$$= \int_{-\infty}^{a}\int_{-\infty}^{b} f_{XY}(x,y)\,dx\,dy = F_{XY}(a,b)$$

$u(a-x)$
ie $u(a-x) \geq 0$
$x \leq a$

RHS.

$$= \int_{-\infty}^{\infty} 1_{(-\infty,a)}(x)\, f_X(x)\,dx \cdot \int_{-\infty}^{\infty} 1_{[-\infty,b]}(y)\, f_Y(y)\,dy$$

$$= \int_{-\infty}^{a} f_X(x)\,dx \int_{-\infty}^{b} f_Y(y)\,dy = F_X(a)\,F_Y(b)$$

$\therefore$ RHS = LHS $\quad \therefore \; f_{XY}(x,y) = F_X(x)\, F_Y(y)$

so X & Y are independent

Next we show if X & Y are independent then

$$E[u(a-X)\,u(b-Y)] = E[u(a-X)] \cdot E[u(b-Y)]$$

$\therefore$ we use definition $\quad F_{XY}(a,b) = F_X(a)\, F_Y(b)$

$$\int_{-\infty}^{a}\int_{-\infty}^{b} f_{XY}(x,y)\,dx\,dy = \int_{-\infty}^{a} f_X(x)\,dx \int_{-\infty}^{b} f_Y(y)\,dy$$

$$\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f_{XY}(x,y) \cdot \mathbb{1}_{(-\infty,a]}(x) \cdot \mathbb{1}_{(-\infty,b]}(y)\,dx\,dy$$

$$= \int_{-\infty}^{\infty} f_X(x)\, \mathbb{1}_{(-\infty,a]}(x)\,dx \int f_Y(y)\, \mathbb{1}_{(-\infty,b]}(y)\,dy$$

or $$\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f_{XY}(x,y) \cdot u(a-x)\,u(b-y)\,dy\,dx$$

$$= \int_{-\infty}^{\infty} f_X(x)\, u(a-x)\,dx \cdot \int_{-\infty}^{\infty} f_Y(y)\, u(b-y)\,dy$$

or $$E[u(a-X)\,u(b-Y)] = E[u(a-X)]\, E[u(b-Y)]$$

$\square$.

7 a $P\{X = 1\} = p$

$P\{X = -1\} = 1 - p$ $\qquad N \sim N(\mu = 0, \sigma^2 = 0.25)$.

$Y = X + N(0, 0.25)$

To find joint probability we'll be conditional prob formula

$\cancel{P(X = j, Y \le y) = P(\{X = j\}) \cdot P(\{Y \le y\} | \{X = j\})}$

$\cancel{\text{for } P(\{X = j\}, \{Y \le y\})}$

$$\therefore P(\{Y \le y\} | \{X = j\}) = \frac{P(\{X = j\} \cap \{Y \le y\})}{P(\{X = j\})}$$

$X = +1$

$$\therefore P(\{X = 1\} \cap \{Y \le y\}) = P(\{Y \le y\} | \{X = 1\}) \cdot \underline{P(\{X = 1\})} \quad (1)$$

$\therefore$ Since $Y = X + N$ $\qquad$ where $N \sim N(\mu = 0, \sigma^2 = 0.25)$

$= j + N$

$\therefore Y | X = j \sim N(j, 0.25)$

$\therefore = 1 + N(0, 0.25)$

$= N(1, 0.25)$

$$\therefore P(\{Y \le y\} | \{X = 1\}) = P(\{j + N \le y\}) = P(N \le y - j)$$

we divide by σ both sides to normalize

$$P\left(\frac{N}{\sigma} \le \frac{y - j}{\sigma}\right) = P\left(Z \le \frac{y - j}{\sigma}\right), \quad \frac{N}{\sigma} \to Z \text{ std normal}$$

$$= \phi\left(\frac{y - j}{\sigma}\right)$$

using (1)

$\therefore$ when $j = 1$, $P(X = 1, Y \le y) = p \cdot \phi\left(\frac{y - 1}{0.5}\right)$ Ans

when $j = -1$

$$P(X = -1, Y \le y) = (1-p)\,\phi\left(\frac{y+1}{0.5}\right)$$ Ans

(b) Marginal pdf of X and Y

x	$P_X(x)$
1	p
-1	$1-p$
otherwise	0

$\therefore \; P_X(x) =$

Next marginal pdf of Y, we sum over X since X is discrete r.v.

$$f_Y(y) = \sum_{j \in \{+1,-1\}} f_{Y|X}(y|j)\, P(X=j)$$

$$= p\, f_{Y|X}(y|X=1) + (-p)\, f_{Y|X}(y|X=-1)$$

$$= \frac{p}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(y-\mu)^2}{2\sigma^2}\right) + (1-p)\,\phi\, \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(y-j)^2}{2\sigma^2}\right)$$

$$= \frac{p}{\sqrt{2\pi(0.25)}} \exp\left(-\frac{(y-1)^2}{0.5}\right) + (1-p)\frac{1}{\sqrt{2\pi(0.25}} \cdot \exp\left(-\frac{(y+1)^2}{0.5}\right)$$

$$= \frac{p}{0.5\sqrt{2\pi}} \exp\left(-\frac{(y-1)^2}{2(0.25)}\right) + \frac{1-p}{0.5\sqrt{2\pi}} \exp\left(-\frac{(y+1)^2}{2(0.5)}\right)$$

Ans

(c) Here we have to compare posterior probability.

$P(X=1 \mid Y>0)$ or $P(X=-1 \mid Y>0)$

or $P(\{X=1\} \mid \{Y>0\}) = \dfrac{P(\{X=+1\} \cap \{Y>0\})}{P(Y \geq 0)}$

$P(\{X=-1\} \mid \{Y>0\}) = \dfrac{P(\{X=-1\} \cap \{Y>0\})}{P(\{Y>0\})}$

$$P(\{X=1\} \cap \{Y>0\}) = P(\{Y>0\} \mid \{X=1\})\, P(\{X=1\})$$

$$= (1 - P(1+N>0))\, p$$

$$= (1 - P(N>-1))\, p$$

$$= \left(1 - P\left(Z > \frac{-1}{0.5}\right)\right) p$$

$$= \left(1 - \phi\left(-\frac{1}{0.5}\right)\right) p$$

$$= (1 - \phi(-2))\, p \quad = 0.997\, p$$

Magic :)

$$P(\{X=-1\} \cap \{Y>0\}) = P(Y>0 \mid X=-1)\, P(X=-1)$$

$$= (1 - P(-1+N>0))\,(1-p)$$

$$= \left(1 - P\left(Z > \frac{1}{0.5}\right)\right)(1-p)$$

$$= (1 - \phi(2))\,(1-p)$$

$\therefore P\{X=+1\} \mid Y>0\} > P(X=-1 \mid Y>0)$

$= 0.023\,(1-p)$

We all knew +1 is answer but.....
we arrived at it one way or another!

(8) $X \sim N(0,1)$, $N \sim N(0, \sigma_N^2)$ (12)

a) $X \perp\!\!\!\perp N$

$$\therefore f_{XN}(x,n) = f_X(x) \cdot f_N(n)$$

(x, σ_N^2)

Also $Y = X + N$

To find conditional pdf of Y given X

$$\cancel{P\{Y \mid X = x\}} = \frac{\cancel{P(\{Y\} \cap \{X = x\})}}{\cancel{P\{X = x\}}}$$

$$f_{Y|X}(y|x) = \frac{f_{XY}(x,y)}{f_X(x)} = \frac{\cancel{f_X(x)} \cdot f_N(x+N)}{\cancel{f_X(x)}}$$

~~total~~ joint

scalar

$$= \frac{\cancel{f_{XY}(\{X = x\} \cap \{X + N\})}}{f_X(x)}$$

$Y = x + N$

$$= \frac{\cancel{f_{XY}(N(x, \sigma_N^2))}}{f_X(x)}$$

$$= f_{Y|X}(y,x) = \frac{1}{\sqrt{2\pi}\,\sigma_N} e^{-\frac{(y-x)^2}{2\sigma_N^2}}$$

(b) joint pdf

$$f_{XY}(x,y) = f_{Y|X}(y|x) \cdot f_X(x)$$

$$= \frac{1}{\sqrt{2\pi}\,\sigma_N} e^{-\frac{(y-x)^2}{2\sigma_N^2}} \cdot \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}}$$

$$= \frac{1}{\sqrt{2\pi}\,\sigma_N} e^{-\frac{x^2}{2} - \frac{(y-x)^2}{2\sigma_N^2}}$$

8 (c) Condition pdf of X given $Y = y$

$$\therefore f_{X|Y}(x, y) = \frac{f_{XY}(x, y)}{f_Y(y)} \rightarrow \text{joint pdf}$$

$$f_Y(y) = \int_{-\infty}^{\infty} f_{XY}(x, y)\, dx$$

$$= \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}\,\sigma_N} e^{-\frac{x^2}{2} - \left(\frac{y-x}{2\sigma_N^2}\right)^2} dx = I$$

$$= 1/\sqrt{2\pi}\,\sigma_N \; e^{-\frac{x^2}{2} - \frac{(y-x)^2}{2\sigma_N^2}}$$

$$\therefore f_{X|Y}(x, y) = \frac{1/\sqrt{2\pi}\,\sigma_N \cdot e^{-x^2/2 - \left(\frac{y-x}{2\sigma_N}\right)^2}}{\int_{-\infty}^{\infty} 1/\sqrt{2\pi}\,\sigma_N \cdot e^{-x^2/2 - \left(\frac{y-x}{2\sigma_N}\right)^2} dx} = I_1 \text{ (say)}$$

Ans

(d) when $Y = y$, $x_0 = g(y)$ that maximise $P(x_0 < X < x_0 + dx \mid Y = y)$. To find x_0

~~from Bayes theorem~~

$$P(x_0 < X < x_0 + dx \mid Y = y) = \int_{x_0}^{x_0 + dx} f_{X|Y}(x|y)\, dx$$

$$\approx f_{X|Y}(x_0 \mid y)\, dx$$

from (c) we will get I_1 which we need maximizing

$$f_{X|Y}(x \mid y) = \frac{1}{\sqrt{2\pi \frac{\sigma_N^2}{1+\sigma_N^2}}} \exp\left(-\frac{\left(x - \frac{1}{1+\sigma_N^2} y\right)^2}{2 \cdot \frac{\sigma_N^2}{1+\sigma_N^2}}\right)$$

exp is max when $e^{\{u\}}$ is 1. so $u = 0$

or

$$x = \frac{\sigma_N}{1+\sigma_N^2}\, y$$

$$\text{or } x_0 = \frac{\sigma_N}{1+\sigma_N^2}\, y$$

9. N time units geometric random variable.

$P_j = (1-a)a^{j-1} \quad \forall j = 1,2,\ldots \qquad \therefore N \sim \text{Geometric}(a)$

$$P(N=n) = (1-a)\, a^{n-1} \qquad n = 1,2\ldots$$

~~pmf of K~~

$$P(K=k \mid N=n) = \binom{n}{k} p^k (1-p)^{n-k}, \quad k = 0,1,\ldots,n$$

pmf of K → the marginal pmf of K

$$P(K=k) = \sum_{n=1}^{\infty} P(K=k \mid N=n)\, P(N=n)$$

$$= \sum_{n=1}^{\infty} \binom{n}{k} p^k (1-p)^{n-k} (1-a)\, a^{n-1}$$

$$= \sum_{n=1}^{k-1} \binom{n}{k} p^k (1-p)^{n-k} (1-a)\, a^{n-1} + \sum_{n=k}^{\infty} \binom{n}{k} p^k (1-p)^{n-k} (1-a)\, a^{n-1} \quad \text{—— (1)}$$

now $\displaystyle\sum_{n=k}^{\infty} \binom{n}{k} p^k (1-p)^{n-k} (1-a)\, a^{n-1}$

$$= (1-a)\, p^k \sum_{n=k}^{\infty} \binom{n}{k} (1-p)^{n-k} a^{n-1}$$

$$= \frac{1-a}{a(1-p)^k}\, p^k \sum_{k}^{\infty} \binom{n}{k} (1-p)^n a^n$$

$$= \frac{a^k (1-a)\, \cancel{(1-p)^k}\, p^k}{a^k a\, (1-p)^k\, \cancel{(1-p)^k}} \sum_{k}^{\infty} \binom{n}{k} (1-p)^n a^n$$

$$= \frac{a^k (1-a)}{a}\, p^k \sum_{k=0}^{\infty} \binom{n}{k} (1-p)^{n-k} a^{n-k}$$

$$= \frac{a^k(1-a)}{a} p^k \left(1-(1-p)a\right)^{-(k+1)}$$

$$\left((-\beta)\right)^{-(k-1)} = \sum_{k}^{\infty} \binom{r}{k} \beta^{n-k}$$

$$= a^k p^k \frac{(1-a)}{a} \frac{1}{(1-a+ap)^{k+1}}$$

here $\beta = (1-p)a$

$$= \frac{a^k p^k}{(1-a+ap)^k} \cdot \frac{(1-a)}{a(1-a+ap)}$$

$$= \left(\frac{ap}{1-a+ap}\right)^k \cdot \frac{1-a}{a} \cdot \frac{1}{1-a(1-p)}$$

~~in ①, first term~~

in ① first terms cancels because $n \geq K$

$$\therefore \quad P(\{k=0\}) = \frac{(1-a)(1-p)}{1-a(1-p)}$$

Ans